Tools Platform

ELIXIR All Hands, 10-12 June 2024, Uppsala, Sweden

Fortifying the ecosystem for research software

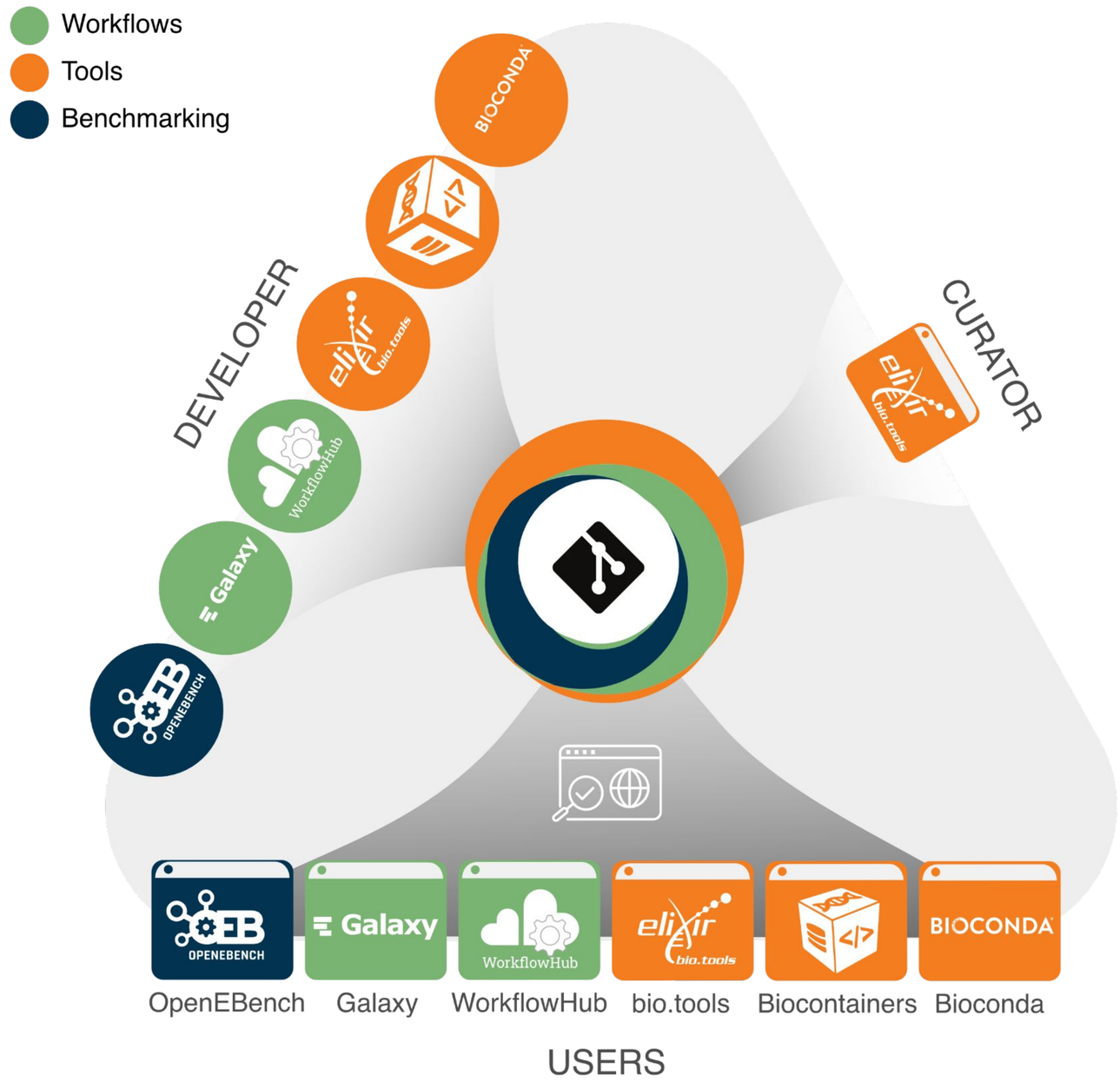
The ELIXIR Tools Platform provides services that enable findability, accessibility, interoperability and reusability (FAIR) for a wide spectrum of **research software**, ranging from individual tools to complex analytical workflows. Beyond research software, the platform also enable the training and evaluation of machine-learning models.

From a **community perspective**, the ELIXIR Tools Platform supports community-driven scientific benchmarking efforts as well as promotes the adoption of software development best practices that should contribute to more sustainable research software as well as the visibility of its developers, **from researchers who code to research software engineers**.

The **Research Software Ecosystem**, a centralised **software metadata repository** represents a transformative approach to addressing the challenges of software metadata accessibility.

By **federating metadata** from diverse sources and adopting a format-agnostic strategy, this *git-based* repository provides a common research software metadata service to all ELIXIR Tools platform associated services.

This ELIXIR initiative holds the potential to significantly enhance the **discoverability**, **usability**, and **reusability** of research software, promoting its efficient use and **making scientific activities more reproducible and replicable**.



research-software-ecosystem.github.io

WP2 The ELIXIR Research Software Ecosystem

Activities
- Designing a **software metadata editor** for the ecosystem
- Defining the next generation of the **Bio.tools search** UI and backend
- **Outreach**
- Expanding the **Ecosystem**

Key expected outcomes
- Consolidation of **Bioschemas**
- **Metadata editor**
- Design new **search interface** for bio.tools

WP3 Software management, stewardship and standards

Activities
- **Best practices** in research software
- **Machine actionability** in Software
- **Quality indicators** for Software
- **Training and capacity building** for Software
- Application of **EDAM** ontology to ELIXIR Communities and diverse ELIXIR Services

Key expected outcomes
- A first version of the ELIXIR *machine-actionable* **software management plan**
- A new version of **EDAM**, with updates for ELIXIR Communities, Services and AI applications

WP4 Defining a roadmap towards reproducible and distributed analysis using benchmarking

Activities
- Defining **standards for benchmarking datasets** across emerging communities
- Facilitating benchmarking through **containerisation**

Key expected outcomes
- Use of **FAIRified benchmarking datasets** from ELIXIR Communities within the OpenEBench ecosystem
- **ARM**-compatible Biocontainers

WP5 Democratised access to reusable complex workflows

Activities
- **Best practices**, upscaling to build FAIR workflows using **WorkflowHub** in Communities
- **Environmental impact** of Galaxy and bioinformatics
- Flexible upscaling of **FAIR workflows**

Key expected outcomes
- New metadata related to **energy consumption**
- Implementation of mitigation strategies to **reduce environmental impact**

Contact

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DOI

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